Matthew J. Cockayne

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RESEARCH INTERESTS

Responsible AI methodologies for healthcare with applications in medical image analysis. Focus on interpretable and fair deep learning through concept bottleneck models, fairness evaluation, bias mitigation, and uncertainty quantification. Expertise in robustness evaluation, clinical validation, and translational research for dermatological and cardiovascular diagnosis. Developing domain-agnostic frameworks for trustworthy AI deployment across medical imaging modalities.

EDUCATION

PhD in Computer Science 2022-2026 (Expected)
Keele University, Keele, UK
Thesis: Responsible AI for Dermatological Prediction
Funding: Competition-funded Department Studentship
Published 2 first-author papers (Pattern Analysis and Applications journal, AIIH 2025 conference oral presentation). Co-author on AI & Ethics journal paper in minor-revisions. Two first-author manuscripts in preparation on concept bottleneck models and fairness-aware learning. PGR Representative for 3 consecutive years.

MSc in Artificial Intelligence and Data Science, Distinction 2021-2022
Keele University, Keele, UK
Data Analyst Internship: Developed explainable AI methods for CV-based sentiment analysis in recruitment. Turing Network Development Research Associate: Organised networking events and hackathons.

BSc in Physics, First Class Honours 2018-2021
Keele University, Keele, UK

RESEARCH EXPERIENCE

PhD Researcher - Responsible AI for Healthcare 2022-Present
Keele University, School of Computer Science and Mathematics
- Developed DermFormer, a transformer-based multi-modal architecture achieving state-of-the-art performance with superior robustness evaluation against real-world corruptions on Derm7pt dataset; validated clinical utility through uncertainty quantification and interpretability analysis
  Code: github.com/xraikeele/DermFormer
- Conducted information-theoretic analysis revealing a majority of diagnostic label information emerges from concept interactions; Implemented interpretable non-linear classifiers for concept bottleneck models maintaining clinical transparency while capturing synergistic relationships
- Implemented fairness-aware curriculum learning reducing performance disparities across Fitzpatrick skin types V-VI, demonstrating translational impact for equitable melanoma detection in underrepresented populations
- Collaborated with clinical partners on cardiovascular disease mortality prediction using multi-center MINAP dataset (400k+ patients), addressing bias mitigation and clinical validation for AI fairness at scale
- Created zero-shot segmentation frameworks combining class activation mapping with foundation models (SAM, MedSAM) for automated dermoscopy analysis; validated on ISIC datasets (35k+ images) demonstrating generalisation
  Code: github.com/xraikeele/CAM-Guided-SAM
- Led reproducible research projects with open-source implementations, comprehensive documentation, and public code releases supporting translational research adoption

Data Analyst Intern - Explainable AI 2021-2022
Industry Partner via Keele University
- Developed explainable AI techniques for NLP models analysing sentiment in recruitment contexts
- Implemented interpretability methods ensuring transparency and fairness in automated decision-making systems

PUBLICATIONS

Journal Articles

Cockayne, M. J., et al. (2025). “DermFormer: Nested Multi-modal Vision Transformers for Robust Skin Cancer Detection.” *Pattern Analysis and Applications*, Springer Nature. DOI: 10.1007/s10044-025-01572-0

Al-Bander, B., et al. “Investigating and Mitigating Bias in Cardiovascular Disease Mortality Predictors for Scaling-up AI Fairness.” *AI & Ethics*, Springer Nature. Status: Minor Revisions.

Conference Papers

Cockayne, M. J., et al. (2025). "Classification-to-Segmentation: Class Activation Mapping for Zero-Shot Skin Lesion Segmentation." *AIIH 2025*, University of Cambridge. [**Oral Presentation**] DOI: 10.1007/978-3-032-00656-1_24

Manuscripts in Preparation

Cockayne, M. J., et al. "Concept Bottleneck Models with Interpretable Non-Linear Classifiers for Improved Concept Synergy Capture."
Cockayne, M. J., et al. "Fairness-Aware Curriculum Learning for Concept Bottleneck Models: Mitigating Bias in Dermatological Diagnosis Across Skin Types."

TECHNICAL SKILLS

Programming: Python, LaTeX, R, SQL
ML Frameworks: PyTorch, TensorFlow, Keras, scikit-learn, Hugging Face Transformers
Computer Vision: OpenCV, torchvision, TIMM, Segment Anything Model (SAM), melanoma detection
Explainable AI: CAM, GradCAM, integrated gradients, attention visualisation, SHAP
Fairness & Bias: Aequitas, Fairlearn, demographic parity, equalised odds, adversarial debiasing
Medical Imaging: MONAI, SimpleITK, medical image preprocessing and augmentation, dermoscopy analysis
Clinical Validation: Uncertainty quantification, robustness evaluation, clinical performance metrics, translational research methodologies
Data Analysis: Pandas, NumPy, SciPy, statistical modeling, information-theoretic analysis
Development: Git/GitHub, Jupyter Notebooks, Weights & Biases, TensorBoard
High Performance Computing: HPC cluster management, SLURM job scheduling, batch script optimization
Datasets: ISIC, Derm7pt, SkinCon, MINAP (cardiovascular risk stratification)

TEACHING EXPERIENCE

Laboratory Demonstrator 2022-Present
Keele University, School of Computer Science and Mathematics
- Supported undergraduate and master's students across computer science modules for 4 years
- Delivered guest lectures on deep learning, computer vision, and responsible AI
- Mentored students on machine learning and AI ethics research projects

PROFESSIONAL SERVICE

Peer Review: Reviewer, *Engineering Applications of Artificial Intelligence* journal (4 reviews, 2024-2025)
Leadership: Postgraduate Research Representative, Keele University (2022-2025, 3 years); Student Representative, Physics Programme (2018-2021, 3 years)
Mentorship: Supervised undergraduate and master's research projects on machine learning and AI ethics; provided guidance on responsible AI methodologies and fairness evaluation techniques
Outreach: Data Science Content Creator Internship (2021-2022) - developed educational resources and materials on AI and data science fundamentals for broader audiences
Community: Turing Network Development Research Associate (2021-2022) - organised networking events, hackathons, and interdisciplinary research initiatives to foster cross-disciplinary collaboration

PRESENTATIONS AND TALKS

Oral Presentation: "Classification-to-Segmentation: Class Activation Mapping for Zero-Shot Skin Lesion Segmentation." AIIH 2025, University of Cambridge (Sep 2025)
Research Seminars: Presented research on responsible AI for medical imaging, concept bottleneck models, and fairness-aware learning at departmental research seminars, Keele University (2022-2025)
Guest Lectures: Deep learning, computer vision for medical imaging, responsible AI (fairness, interpretability, bias mitigation), transformers in healthcare (2022-2025)

AWARDS AND HONORS

Competition-funded PhD Departmental Studentship (2022) | MSc Distinction (2022) | First Class Honours BSc (2021) | Turing Network Development Research Associate (2021-2022)

REFERENCES

Available upon request